

A.4 Figure of Modular Design of Agents in FINCON

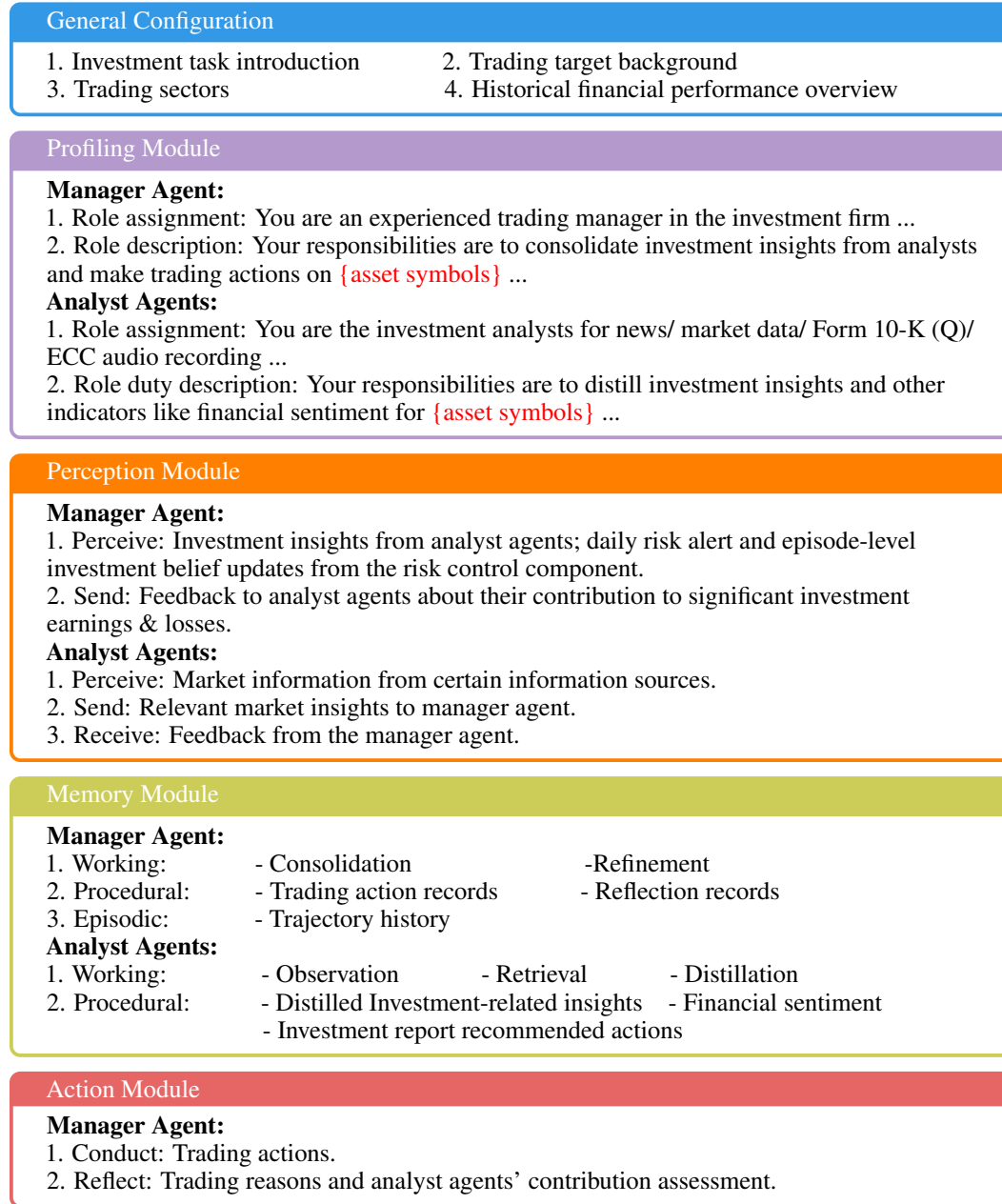


Figure 4: The detailed modular design of the manager and analyst agents. The general configuration and profiling modules generate text-based queries to retrieve investment-related information from the agents' memory databases. The perceptual and memory modules interact with LLMs via prompts to extract key investment insights. The action module of the manager agent consolidates these insights to facilitate informed trading decisions.

A.5 Experimental Setup

Multi-Modal Datasets. We collect a comprehensive multi-modal dataset to simulate a realistic market environment. This dataset includes stock price data, daily financial news, company filing reports (10K and 10Q), and ECC (Earnings Call Conference) audio recordings spanning from January 3, 2022, to June 10, 2023. Each data source is assigned to specific analyst agents based on the timeliness of the information. For example, annual filings (10K) exhibit longer-term persistence, quarterly filings (10Q) and ECC data have medium-term relevance, and daily financial news provides the most immediate information.

Evaluation Metrics. We evaluate FINCON and benchmark it against other state-of-the-art LLM-based and DRL-based agent systems using three key financial performance metrics: Cumulative Return (CR%) [90], Sharpe Ratio (SR) [91], and Max Drawdown (MDD%) [92]. These metrics help quantify each model’s profitability, risk-adjusted returns, and risk management performance, respectively.

Comparative Methods. In the single-stock trading task, we compare FINCON against seven algorithmic agents and the widely accepted Buy-and-Hold (B & H) baseline. The three DRL-based agents—A2C, PPO, and DQN—are from the FinRL framework [48], while the four state-of-the-art LLM-based agents include GENERATIVE AGENT [20], FINGPT [93], FINMEM [32], and FINAGENT [33]. For portfolio management, we benchmark FINCON against the classical Markowitz MV portfolio selection strategy [1], the RL-based FinRL-A2C agent [48], and the B & H strategy, which holds an equal-weighted position across all assets (equal-weighted ETF). Our focus on classical, RL-based, and B & H methods is due to the current lack of mature LLM-based agents for portfolio management tasks.

Implementation Details. In our experiments, all LLM-based agent systems, including FINCON, use GPT-4-Turbo as the backbone model, with the temperature parameter set at 0.3 to balance response consistency with creative reasoning. FINCON is trained on financial data from January 3, 2022, to October 4, 2022, and tested on data from October 5, 2022, to June 10, 2023. Since deep reinforcement learning (DRL) agents require extensive data for convergence, their training period is extended to nearly five years (January 1, 2018, to October 4, 2022) to ensure fair comparison. The testing period remains the same across all models. The final performance metrics are based on the test trajectory with the median CR and SR values from five repeated epochs. If the median CR and SR occur in different epochs, performance is assessed based on the trajectory with the median CR value.

A.6 Single Stock Trading Result Graphs



Figure 5: CRs over time for single-asset trading tasks. FINCON outperformed other comparative strategies, achieving the highest CRs across all six stocks by the end of the testing period, regardless of market conditions.

A.7 Detailed Ablation Study